

Reconfiguring Agency in the Age of Algorithmic Control

An Interactive Performance Lecture

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Abstract

Where is our agency in the age of algorithms? This research investigates how algorithmic systems use the **illusion of agency** as a means of control, treating human bodies as both working machines and sites for data extraction. The bodies are simultaneously reduced into data points and multiplied through digital identity labor, resulting in **splintered selves** and **fragmented agency**. With the rise of AI agents, human agency faces a deeper crisis. Feminist, or Cyberfeminist, theories, in particular Lucy Suchman's work on feminist reconfigurations in technology design, provide a different perspective: agency is always in flux—relational, contextual, and reconfigurable.

To practice **reconfigured agency**, this research further takes the form of an experimental **performance lecture** using interactive technology. During a 15-minute performance, audience participants gain real-time control of the artist (the author) and her digital avatars via a custom web-based application, while the artist recites a speech on human agency, accompanied by layered audio-visual effects. This format demonstrates the tension between individual autonomy and systemic control while enacting a shared, fluid agency, emerging from the continuous, dynamic interactions.

CC Concepts

• Human-centered computing → Human computer interaction (HCI) • Human-centered computing → HCI theory, concepts and models • Human-centered computing → Interaction design process and methods • Human-centered computing → Collaborative and social computing theory, concepts and

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UbiComp Companion '25, Espoo, Finland

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ACM ISBN 979-8-4007-1477-1/25/10.

<https://doi.org/10.1145/3714394.3756261>

paradigms • Computer systems organization → Real-time systems • Social and professional topics → Gender • Social and professional topics → Surveillance • Theory of computation → Design and analysis of algorithms

Keywords

Agency, Algorithmic Control, Technology & Society, Cyberfeminism, Tech Feminism, Posthumanism, Human-Computer Interaction, Interactive Technology Design, System Design, Interactive Media Arts, Performance Lecture, Performance Art, Creative Technology

ACM Reference format:

Mengna Da. 2025. Reconfiguring Agency in the Age of Algorithmic Control: An Interactive Performance. In *Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '25)*, October 12-16, 2025, Espoo, Finland. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3714394.3756261>

The rest of the paper will be divided into five sections. The first three sections summarize the research, corresponding to the three chapters of the performance: THE GHOST, THE BODY, and THE RECONFIGURATION. The last two sections provide more technical details about the performance lecture including its system design and performance progression.

1 Illusion of Agency as A Means of Control (THE GHOST)

Agency, the fantasy of autonomy, the lucid dream of control. Agency is usually defined as the capacity to think freely, to act on one's will, and to exercise control. But in the age of algorithms, the notion of agency becomes deeply complicated. Actions become reactions—filtered and shaped by algorithms that are programmed to reinforce our sense of control. For example, as we scroll through infinite feeds on social media, the illusion of choice obscures the deeper structure: it is not us choosing what we see, but by algorithms selecting what is most likely to keep us engaged. Each click or swipe appears to be an autonomous decision, but in fact reflects a behavioral loop engineered through data extraction, surveillance, and prediction.

Researcher and artist Vladan Joler describes our reality as a *Platopticon*¹—combining Plato’s Allegory of the Cave, which framed perception as illusion, and Jeremy Bentham’s Panopticon, a design for an omnipresent surveillance prison. In Joler’s *Platopticon* model, the data center in the middle serves both as the powerhouse and the control center. Each of us inhabits a digital cell-cave, filled with content tailored to our tracked behavior, curated by systems beyond our reach.

Humans crave freedom and agency. Today, the most effective means of control is not overt coercion but the illusion of agency. Each individual in this system is constantly rewarded with micro feedback loops that reinforce a sense of control, even when the system shapes and limits the field of possible actions. The more personalized the experience, the more it traps us. This is not a loss of agency as disappearance, but as appropriation: control displaced and mirrored back to us through interfaces designed for engagement, not emancipation.

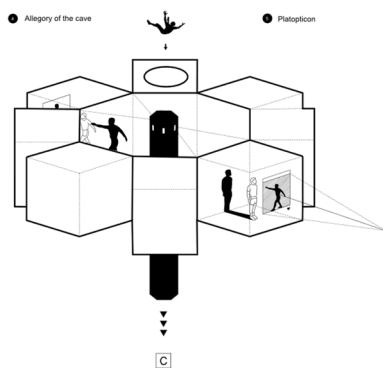


Figure 1. Vladan Joler, “New Extractivism”, extractivism.online, 2020. ©Vladan Joler

2 Datafied Body, Digital Labor, and Splintered Selves (*THE BODY*)

In Joler’s *Platopticon* metaphor, each body becomes simultaneously a resource, a product, and a worker.² Every piece of data about our bodies is captured to render a multidimensional profile, which is in turn used to design content that impacts our actions. Everything we say and do becomes unpaid labor that feeds the machine and helps train the model.

History further contextualizes this dynamic. The modern perception of the body as a working machine originates from the so-called Age of Enlightenment. Marxist feminist Silvia Federici argues that early scientific developments and the mechanical philosophy during this era did not lead to liberation but rather contributed to the notion of the body as an instrument for labor

and reproduction, separated from the intellect.³ In particular, colonized people and women’s bodies were and still are exploited as if they are public territories, as evidenced by slave trades and witch hunts.

The same logic persists today, now extended into the digital domain. Our data is extracted and monetized while our existence remains invisible, except as digital traces and data points. This extractive model perpetuates societal biases and problems. Those who do not fit within mainstream categories—such as queer people, racial minorities, and disabled people—are often misclassified and even erased as outliers. Algorithmic systems trained on biased data reproduce and reinforce existing social inequalities.

On the other hand, we are coerced to produce and maintain multiple digital identities. This type of digital identity labor is essential for freelancers or even regular employees to navigate today’s reputation economy, in which everyone’s online profiles become their resume. As Kristian Lukić points out, it is done “in similar fashion like sex workers in windows of red light districts.”⁴

The result of all the above has multiple layers: our bodies are simultaneously *flattened* into data and *multiplied* into an array of digital identities, eventually producing a digital dissonance and a sense of splintered selves. Agency is fragmented.



Figure 2. Screenshot from the artist’s video essay accompany the performance lecture, illustrating a flattened and multiplied individual. ©Mengna Da

3 Reimagining Agency in the Age of AI (*THE RECONFIGURATION*)

With the increasing adoption of Artificial Intelligence (AI), in particular “AI agents,” we confront a new rhetorical and material frontier. IBM prescribes that an AI agent is a system or program that is capable of performing tasks independently. AI agents are envisioned to perceive the environment, reason, and act (ReAct) to perform tasks autonomously. However, the marketing of the allegedly autonomous AI agents and tools masks the immense human labor that sustains them, ranging from users who,

¹ Vladan Joler. 2020. *New Extractivism*. Retrieved from <https://extractivism.online/>

² Joler.

³ Silvia Federici. 2004. *Caliban and the Witch: Women, the Body and Primitive Accumulation*. Autonomedia, New York.

⁴ Kristian Lukić. 2016. *Colonization with love*. Share Lab. Retrieved from <https://labs.rs/en/colonization-with-love/>

willingly or not, provide data to train the machine models to those who build physical chips and infrastructures. This erasure of human input and labor renders humans as ghosts in the algorithms, essential but invisible.

Science and technology researcher and anthropologist Lucy Suchman offers a critical alternative. Suchman argues that agency is never a private property; it is always relational and contextual.⁵ As Karen Barad writes, quoted by Suchman, “Agency is not an attribute but the ongoing reconfigurings of the world.”⁶ In complex sociotechnical systems, such as medical procedures—or an interactive performance, which the paper will discuss in more detail below—agency is distributed through ongoing interactions of various participants in the system, whether human or nonhuman or, in old-fashioned computing terms, “master” or “servant.” Suchman cites anesthesia as an example: during anesthesia, the patient’s agency is not simply transferred from patient to doctor but is delegated across an interconnected system of machines, medicines, nurses, doctors, and the patient. Agency is not a private property to hand off or be taken away. Instead, agency is always being reconfigured, tied to the specific sociotechnical arrangements we are part of.

What challenges human agency is not the algorithm itself, but the false belief that the algorithm can be wielded as a weapon for one group to control another. This paper proposes that we must recognize agency as collective, relational, and reconfigurable. We are all impacted, all essential, and all responsible—we must imagine, experiment with, and practice a new type of reconfigured agency to challenge and break the current algorithmic control.

4 An Interactive System Designed to Explore Reconfigured Agency

The artist’s interactive performance lecture builds on the Cyberfeminist theory of *reconfigured agency*. During the live performance, viewers can control the artist’s bodily actions via a real-time participatory application on the web, and the artist follows their instructions as if she were a humanoid while simultaneously giving a lecture on agency.

The system is designed to delegate the artist’s agency to the audience but mediated through algorithms. The audience is provided with a QR code, which leads them to a custom web-socket application that can receive real-time input from the audience; their selections are then processed through a voting system, and the top choice will be announced by a robotic voice every 15 seconds unless another option receives more votes during the 15-second interval.



Figure 3. Live performance lecture at New York University Shanghai in June, 2025. ©Mengna Da

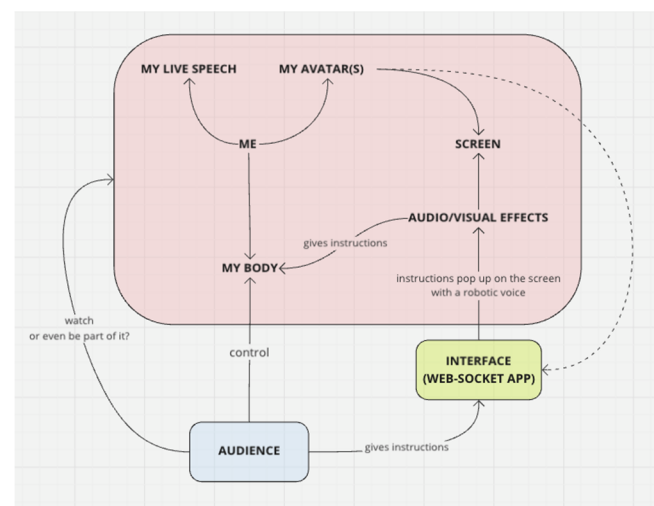


Figure 4. System design diagram by the artist. ©Mengna Da

In the last chapter, digital avatars of the artist start to appear in the background. The digital avatars are created with 3D scanning, MotionCapture (MoCap), Mixamo animations, and Three.js. The artist breaks from the previous role and stops following instructions; instead, she delegates the tasks—the audience voted actions—to her digital avatars. As she lectures about reconfigured and distributed agency, the voting algorithm would be replaced by a different system in which each selection would immediately trigger an action on one of the avatars. This design raises question about where agency lies: is it with the audience, or the system designer (the artist), or the system itself?

⁵ Lucy A. Suchman. 2016. “Agencies in technology design: Feminist reconfigurations.” In *Machine Ethics and Robot Ethics*, Wendell Wallach and Peter Asaro (Eds.). Routledge, London, 361–375. Retrieved from <https://www.lancaster.ac.uk/fass/resources/sociology-online-papers/papers/suchman-agenciestechdesign.pdf>

⁶ Karen Barad. 2003. Posthumanist performativity: Toward an understanding of how matter comes to matter. *Signs: Journal of Women in Culture and Society*, 28, 3 (2003), 801–831. <https://doi.org/10.1086/345321>

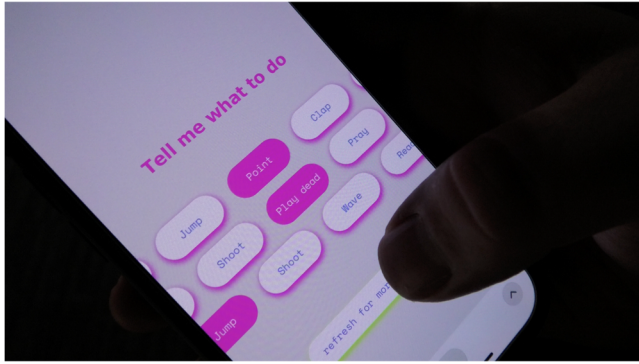


Figure 5. Custom web-socket app on the mobile. ©Mengna Da

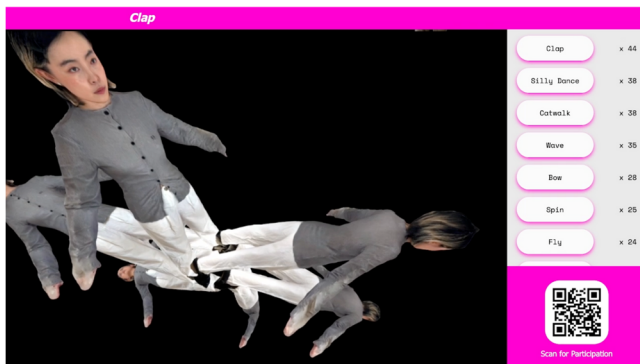


Figure 6. Screenshot of the web-socket app on the presenter's end, with live audience votes on the right and pre-made animations on the left. ©Mengna Da

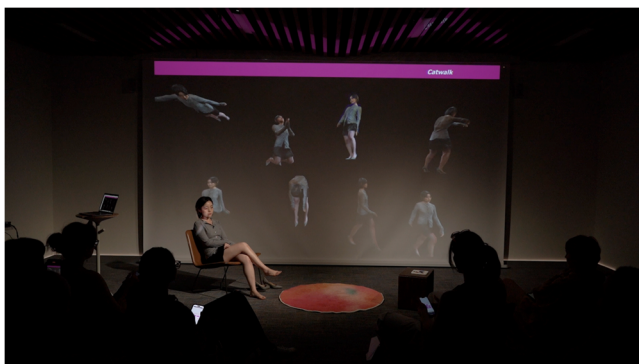


Figure 7. Live performance lecture at New York University Shanghai in June, 2025.. ©Mengna Da

The performance lasts 15 minutes and unfolds in three chapters—THE GHOST, THE BODY, and THE RECONFIGURATION—each blending live participation, performance, lecture, and video essay in a way that takes the audience on a journey from simple interactions to layered audiovisuals, culminating in an overload of the artist's digital avatars.

5 (Non-)Conclusion and Future Iterations

This performance creates a system of control, and yet the control is not simply one-way—the artist, the participants, and all the technologies involved are dependent on each other. The artist

would not have completed the piece without the audience interaction, and the audience wouldn't have the power to control the artist without the system she designed. It makes visible how agency emerges from the continuous, dynamic interactions.

The performance has taken place both online and in person several times with different audience sizes, ranging from under ten participants to a full house of nearly one-hundred people. Based on post-performance conversations and casual interviews, different patterns of interaction were identified: some people showed a competitive trait of wanting to control the voting results and therefore interacting with the app constantly, while others felt uncomfortable directing the artist's body especially when there are only a few participants; the more participants there were, the more anonymous it felt to certain people; more tech-oriented viewers spent most of the time trying to figure out technical details, while some people focused more on the lecture.

For future iterations, the artist hopes to conduct more formal surveys to collect feedback. The artist also intends to experiment with different modes of interaction—for example, what if there is only one device so that the participants would send instructions one at a time and therefore take on more responsibility for their choice? Or instead of a voting system, are there other systems that illustrates the idea of reconfigured agency?

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